$F_1$ catalytic unit: Three α and three β chains combine to form hexameric $\alpha_3\beta_3$ that is responsible for catalyzing the synthesis of ATP. The γ and ε polypeptide chains organize to form the central stalk and run to the inner cavity of the hexameric ring. The γ subunit helps in holding the $\alpha_3\beta_3$ and manifests rotating of the machine. This complex harness energy released by the electron transport to drive the synthesis of ATP, whereas, while alone, the $F_1$ component hydrolyses ATP.

$F_0$ region of proton pump: It consists of 10-14 c subunits organized into a ring-like structure and acts as a proton channel. A single 'c' unit connects $F_0$ to $F_1$ unit. $F_0$ and $F_1$ are connected at two points through γε central stalk and through the arm formed by 'a' subunit, two b subunits and one δ subunit.

### 8.4.2.2 Free energy of hydrolysis of ATP

Among phosphate-containing molecules, ATP is an important molecule. It exchanges energy between enzymes. ATP is an unstable molecule which hydrolyzes to ADP and inorganic phosphate. The high energy of this molecule comes from the two high-energy phosphoanhydride bonds.

The hydrolysis of ATP has a $\Delta G^{\circ\prime}$ of –30.5 kJ/mol. In addition, since the cells contain higher concentration of ATP than ADP, the ΔG for the reaction is more negative than the $\Delta G^{\circ\prime}$ value. ATP hydrolysis can therefore donate energy to other systems to allow those systems to perform reactions that would otherwise be thermodynamically unfavorable.

The phosphate bond holds usable energy. The hydrolysis of a phosphate which is responsible for release of free adenosine from AMP has a $\Delta G^{\circ\prime}$ of –14 kJ/mol; this is not considered to be a high-energy phosphate bond; and is emphasized by drawing the bond as a straight line.

The net result is that ATP hydrolysis can act as a thermodynamic driving force to allow thermodynamically unfavorable reactions that do not involve phosphate transfer to proceed. Though ATP is the most widely used high-energy compound, there are other high-energy compounds of physiological significance. An example is the high-energy thioester bond of acetyl-CoA.

## 8.5 UNCOUPLERS

Chemicals such as 2,4 dinitrophenol (DNP) act as uncoupling agents, which means they stop the synthesis of ATP, but ETC still continues and so oxygen is still consumed. DNP and other uncouplers are lipid soluble small moieties and can bind $H^+$ ions and transport them across membranes.



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